

McNemar's Test for Paired Data

Presentation Outline

- 1 Variables and Data
- 2 McNemar's Test
- 3 McNemar's Confidence Interval

Variables

- Variable 1
 - $I = 2$ categories
- Variable 2
 - $J = 2$ categories
- Similar variables with same categories
 - Ex. Success/Failure, Yes/No, Correct/Incorrect, etc.
 - Variables are matched or related in some way.

Sample Data Table

Data Table contains n rows with observations from the two categorical variables.

Variable 1	Variable 2
Success	Failure
Success	Failure
Failure	Success
⋮	⋮
⋮	⋮
Failure	Success
Success	Success

Contingency Table

Form contingency table for two variables by cross classifying observations.

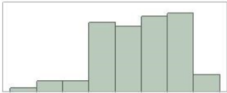
Variable 1	Variable 2		Total
	Success	Failure	
Success	Y_{11}	Y_{12}	$Y_{1.}$
Failure	Y_{21}	Y_{22}	$Y_{2.}$
Total	$Y_{.1}$	$Y_{.2}$	n

Example - Histogram Questions

Two related questions about the same histogram were presented to all students in an introductory statistics course.

Each question is scored as either Correct or Incorrect.

A histogram for the sale price of 100 cars is pictured below.



Question A3: Choose the best description of the horizontal axis for the histogram.	Question A4: Choose the best description of the vertical axis for the histogram.
a. The number of cars	a. <i>The number of cars</i>
b. <i>The sale prices</i>	b. The sale prices
c. The mean sales price for each bin	c. The mean sales price for each bin
d. The median sales price for each bin	d. The median sales price for each bin

Example - Contingency Table

589 students answered the two questions. Their scored responses are summarized in this contingency table.

Question 1	Question 2		Total
	Correct	Incorrect	
Correct	469	12	481
Incorrect	30	78	108
Total	499	90	589

Comparing Marginal Proportions

Proportion of Success on each Variable

- Variable 1: $p_{1.}$
- Variable 2: $p_{.1}$

Are these two proportions the same?

- $p_{1.} = p_{11} + p_{12}$
- $p_{.1} = p_{11} + p_{21}$

If $p_{1.} = p_{.1}$, then we must have that $p_{12} = p_{21}$

Variable 1	Variable 2		Total
	Success	Failure	
Success	p_{11}	p_{12}	$p_{1.}$
Failure	p_{21}	p_{22}	$p_{2.}$
Total	$p_{.1}$	$p_{.2}$	1

McNemar's Test - Hypotheses

Null Hypothesis

- Proportion of success on Variable 1 is same as proportion of success on Variable 2.
- $H_0 : p_{1.} = p_{.1}$

Alternative Hypothesis

- Proportion of success on Variable 1 is different from proportion of success on Variable 2.
- $H_0 : p_{1.} \neq p_{.1}$

McNemar's Test - Test Statistic

If H_0 is true, we also have $p_{12} = p_{21}$.

We will need to compare these estimates $\hat{p}_{12} = \frac{Y_{12}}{n}$ and $\hat{p}_{21} = \frac{Y_{21}}{n}$. This is the same as comparing Y_{12} to Y_{21}

Large differences between Y_{12} to Y_{21} indicate $p_{12} \neq p_{21}$ which implies $p_{1.} \neq p_{.1}$

Test Statistic:

$$z^2 = \frac{(Y_{12} - Y_{21})^2}{Y_{12} + Y_{21}}$$

Large values of z^2 indicate $p_{12} \neq p_{21}$ which implies $p_{1.} \neq p_{.1}$

McNemar's Test - P-value

As long as $Y_{12} + Y_{21} \geq 20$, the distribution of z^2 can be well approximated by the χ_1^2 distribution.

P-Value:

$$P(\chi_1^2 > z^2)$$

Example - McNemar's Test

Below is the contingency table of student responses to the two questions about a histogram.

Question 1	Question 2		Total
	Correct	Incorrect	
Correct	469	12	481
Incorrect	30	78	108
Total	499	90	589

Example - McNemar's Test

Null Hypothesis

- The proportion of correct responses on Question 1 is the same as the proportion of correct responses on Question 2.
- $H_0 : p_{1.} = p_{.1}$

Alternative Hypothesis

- The proportion of correct responses on Question 1 is different from the proportion of correct responses on Question 2.
- $H_0 : p_{1.} \neq p_{.1}$

Example - McNemar's Test

Test Statistic:

$$z^2 = \frac{(Y_{12} - Y_{21})^2}{Y_{12} + Y_{21}} = \frac{(12 - 30)^2}{12 + 30} = 7.714$$

P-value:

$$P(\chi^2 > 7.714) = 0.0055$$

Conclusion: We have very strong evidence of a difference in the proportion of students who correctly answer both questions.

Confidence Interval for Difference in Marginal Proportions

Goal: Confidence Interval for $p_{1.} - p_{.1}$

- Estimate:

$$\hat{p}_{1.} - \hat{p}_{.1} = \frac{Y_{1.}}{n} - \frac{Y_{.1}}{n}$$

- Standard Error of Estimate:

$$SE(\hat{p}_{1.} - \hat{p}_{.1}) = \frac{\sqrt{(Y_{12} + Y_{21}) - (Y_{12} - Y_{21})^2/n}}{n}$$

Confidence Interval for Difference in Marginal Proportions

- Distribution:
As long as $Y_{12} + Y_{21} \geq 20$, the distribution of the estimate $\hat{p}_{1.} - \hat{p}_{.1}$ is well-approximated by the standard Normal distribution Z .
- Confidence Interval:

$$\hat{p}_{1.} - \hat{p}_{.1} \pm z_{1-\frac{\alpha}{2}} SE(\hat{p}_{1.} - \hat{p}_{.1})$$

where $z_{1-\frac{\alpha}{2}}$ is the $1 - \frac{\alpha}{2}$ percentile of the standard Normal distribution.

Example - Confidence Interval for Difference in Marginal Proportions

- Estimate:

$$\hat{p}_{1.} - \hat{p}_{.1} = \frac{481}{589} - \frac{499}{589} = \frac{-18}{589} = -0.0306$$

- Standard Error of Estimate:

$$SE(\hat{p}_{1.} - \hat{p}_{.1}) = \frac{\sqrt{(12 + 30) - (12 - 18)^2/589}}{589} = 0.0109$$

- 95% Confidence Interval

$$-0.0306 \pm 1.96(0.0109) = (-0.0520, -0.0092)$$

Example - Confidence Interval for Difference in Marginal Proportions

Interpretation:

We are 95% confident the proportion of all STAT 101 students who answer correctly when asked to identify the horizontal axis in a histogram is between 0.0092 and 0.0520 lower than the proportion of all STAT 101 students who answer correctly when asked to identify the vertical axis in a histogram.

Relationship to Other Methods

- Matched Pairs

Similar situation but variables are quantitative/numerical. Comparison involves means.

- Difference of Two Proportions

The proportions are for the same variable from two different groups/populations.